



M004 Week 4 — English Worksheet

Topic: Player Events · **Course:** Functions & Games · **Time:** about 45 minutes

This week your code reacts to **you**, the player. An **event** like `on player walk:` runs by itself — no chat command needed.

1 · Predict 🧠

```
on player walk:
    if block below is gold:
        say "Treasure room!"
```

The player walks across stone, stone, gold, stone. When does the message appear? How many times?


```
on player place block:
    say "You built something!"
```

The player places 3 blocks, then walks around for a while. How many times does the message appear? Does walking trigger it?


```
define drop_reward():
    place diamond block

on player walk:
    if block below is redstone:
        drop_reward()
```

The player steps onto a redstone block. What happens, step by step? What happens on normal ground?

2 · Spot the Bug 🐛

Each block below is broken. Rewrite it so it works, then explain why the original was wrong and why your fix works.

Bug A — A trap message should appear **by itself** whenever the player walks onto gold. But nothing ever happens unless someone types `trap` in chat.

```
on chat command "trap":
    if block below is gold:
        say "Trap!"
```

Hint: this should be an **event** that runs on its own, not a chat command.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The player should get **one** welcome reward when they first step onto the redstone start tile. But the reward keeps firing again and again on every step, flooding the world with diamonds.

```
on player walk:
    if block below is redstone:
        place diamond block
```

Hint: use a variable like `reward_given` to remember if the reward already happened.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The message should appear when the player **places a block**, to celebrate building. But right now it appears every time the player simply walks.

```
on player walk:  
  say "Nice building!"
```

Hint: the behavior is right, the **event** is wrong.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Build a mini game zone with **one trap** and **one reward**, each triggered by a player event (**on player walk:** or **on player place block:**).

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.